

Jay Van Vliet

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Professional Experience

iRhythm Technologies, Inc.

San Francisco, CA

Sr. Electrical Test Engineer II

Apr 2021 - Present

- Serve as SME for medical device standards, including IEC 60601 series, EC12, ANSI C63.27, and DO-160.
- Own and drive electrical team process improvement and bi-weekly meeting.
- Leading electrical subsystem design verification strategy for current product cycle.
- Served as electrical test SME for four 510k submissions, resolving all FDA additional information requests.
- Automated electrical design verifications tests in python reducing test and data analysis time by 50% and eliminating human error.
- Developed 50+ electrical subsystem test cases, authored supporting protocols and reports.
- Acted as electrical SME and action owner for two corrective and preventative action plans.
- Led FCC Title 47 certification across two product lines.
- Authored two gap analysis reports pre-verification to validate standards conformance and mitigate regulatory risk.
- Designed three test tool PCBAs to streamline manual test procedures and enable modular, automated test setups.
- Designed breakout PCBA for next-gen vitals monitor to enable prototype product bring up.
- Owned electrical design of manufacturing fixture for high-speed USB data extraction via pogo-pin interface.
- Configured electrical teams Altium 365 workspace environment and component library and guided migration of PCBA designs and component library from Bitbucket.
- Recruited, mentored, and managed electrical engineering interns from January 2022 to present.

Sr. Electrical Test Engineer I

Aug 2018 - Apr 2021

- Collaborated with design engineers on component selection and design for test strategies for next-gen ECG monitor.
- Worked with electrical design engineer to develop, execute and document bring-up tests for next-gen ECG monitor.
- Debugged next-gen product electrical design including critical ECG filter.
- Designed breakout PCBA for next-gen ECG monitor, serving as a versatile tool for prototype bring-up, design verification, and field failure analysis across the organization.
- Designed automation test fixture, including equipment selection and custom interface PCBA design, for firmware automated test team based on fixture requirements significantly reducing manual test burden.
- Developed 20+ electrical subsystem test cases, authored supporting protocols and reports, and executed testing for 510k submission.
- Reviewed electrical designs and provided direct input during the development of an 8-up manufacturing tester for a PCBA used in medical devices reaching millions of patients annually.
- Partnered with manufacturing and software engineering teams to define, develop, and validate the automated test sequences executed by the 8-up production tester.

Element Science, Inc.

San Francisco, CA

Electrical Test Engineer

Oct 2016 - Aug 2018

- Developed Automated Test Equipment (ATE) hardware including PCBA schematic creation and peripheral equipment selection. ATE used for electrical design verification, firmware automated and manufacturing test.
- Wrote test design document and Python automation for electrical hardware design verification and manufacturing electrical hardware test.
- Developed manufacturing electrical test limits based on product specifications and characterization data.
- Performed initial functional test and debug of product PCBAs and test PCBAs.
- Performed characterization of critical microvolt and high voltage product PCBA circuits.

Texas Instruments, Inc.

Santa Clara, CA

Silicon Validation Engineer - ADAS & Infotainment Serializes/Deserializers

Sept 2015 - Oct 2016

- Created schematic for silicon validation PCB and silicon reliability PCBs.
- Built and automated bench tests, correlating bench data to production ATE results to verify effective part screening, and datasheet specification accuracy.
- Worked with a team to analyze customer return ICs and customer systems to optimize production test coverage to near zero failures at customer site..
- Developed and maintained production yield tracking dashboard for 10+ ICs, automating data visualization and reducing time to prepare yield analysis for management from hours to minutes.
- Developed a binary search routine, mitigating exponential test time growth across new product development.

Product and Test Engineer (Rotation)

Dallas, TX; Portland, ME; Baguio, Philippines

Jun 2013 - Sept 2015

- Test Engineering Position – Mixed Signal Automotive Sensors: Deployed ATE code for wafer/IC solutions to recover a delayed project; executed GR&R testing and post-release yield improvements.

- Design Engineering Position – Power Management ICs: Executed performance testing and component investigations to support new product designs.
- Wafer Fabrication Product Development Engineering Position – Power and Legacy Devices: Coordinated process conversion, saving \$1.5M in equipment; collaborated with manufacturing to reduce yield loss.

Education

University of Cincinnati	Cincinnati, OH
Bachelor of Science in Electrical Engineering	April 2013

Technical Skills	Process and Compliance	Tools and Equipment
• Python, pytest, MicroPython, CircuitPython, SCPI • Altium Designer • LTSpice	• IEC 60601-1 and collateral standards (1, 1-2, 4-2, 2-27, 2-4) • DO-160 • EC12 • ANSI 63.27 • FCC Title 47	• Oscilloscopes, DMM, power analyzer, spectrum analyzer, DAQ (LabJack)

Recent Projects

Alarm Clock Modification - reverse-engineered alarm clock design, and modified it to integrate 555 timer and a load switch to convert an "always-on" display to a light that activates upon pressing the snooze button.

Automated Chicken Coop Door - Built an automated door system using an RP2040 microcontroller and CircuitPython. Integrated an RTC for precise timing, a linear actuator with motor driver, and a boost regulator to manage motor power delivery.